# *ANSWERS*

*1.*

**(a) (i)** angle subtended at centre of circle ........................................................................B1
arc equal in length to the radius ...................................................................................B1

**(ii)** arc = $r\theta$ and for one revolution, arc = $2\pi r$ ............................................................M1
so, $\theta = 2\pi r/r = 2\pi$ ..........................................................................................A0

**(b) (i)** *either* weight provides/equals the centripetal force
*or* acceleration of free fall is centripetal acceleration ....................................B1
$9.8 = 0.13 \times \omega^2$ ..........................................................................................M1
$\omega = 8.7$ rad $s^{-1}$ ..........................................................................................A0

**(ii)** force in cord = weight + centripetal force (*can be an equation*) ..........................C1
force in cord = $(L - 13) \times 5/1.8$ *or* force constant = 5.0/1.8 ...............................C1
$(L - 13) \times 5/1.8 = 5.0 + 5/9.8 \times L \times 10^{-2} \times 8.7^2$ ..................................................C1
$L = 17.2$ cm ..........................................................................................A1
(*constant centripetal force of 5.0 N gives L = 16.6 cm allow 2/4*)

*2.*

| | | | |
|---|---|---|---|
| **(a)** | **(i)** | angle (subtended) at centre of circle | B1 |
| | | by an arc equal in length to the radius (of the circle) | B1 |
| | **(ii)** | angle swept out per unit time / rate of change of angle | M1 |
| | | by the string | A1 |

| | | |
|---|---|---|
| **(b)** | friction provides / equals the centripetal force | B1 |
| | $0.72\,W = md\omega^2$ | C1 |
| | $0.72\,mg = m \times 0.35\omega^2$ | |
| | $\omega = 4.49$ (rad $s^{-1}$) | C1 |
| | $n = (\omega/2\pi) \times 60$ | B1 |
| | $= 43\ \text{min}^{-1}$ (allow 42) | A1 |

| | | |
|---|---|---|
| **(c)** | *either* <u>centripetal</u> force increases as $r$ increases | |
| | *or* <u>centripetal</u> force larger at edge | M1 |
| | so flies off at edge first | A1 |
| | *($F = mr\omega^2$ so edge first – treat as special case and allow one mark)* | |

*3.*

**(a)** **(i)** $F = GMm / R^2$ B1

**(ii)** $F = mR\omega^2$ B1

**(iii)** reaction force = $GMm / R^2 - mR\omega^2$ *(allow e.c.f.)* B1

**(b)** **(i)** *either* value of $R$ in expression $R\omega^2$ varies
*or* $mR\omega^2$ no longer parallel to $GMm / R^2$ / normal to surface B1
becomes smaller as object approaches a pole / is zero at pole B1

**(ii)** **1.** acceleration = $6.4 \times 10^6 \times (2\pi / \{8.6 \times 10^4\})^2$ C1
$= 0.034$ m s$^{-2}$ A1
**2.** acceleration = 0 A1

**(c)** e.g. 'radius' of planet <u>varies</u>
density of planet <u>not constant</u>
planet spinning
nearby planets / stars
(*any sensible comments, 1 mark each, maximum 2)* B2

*4.*

**(a)** force per unit mass *(ratio idea essential)* B1

**(b)** $g = GM / R^2$ C1
$8.6 \times (0.6 \times 10^7)^2 = M \times 6.67 \times 10^{-11}$ C1
$M = 4.6 \times 10^{24}$ kg A1

**(c)** **(i)** *either* potential decreases as distance from planet decreases
*or* potential zero at infinity and X is closer to zero
*or* potential $\alpha -1/r$ and Y more negative M1
so point Y is closer to planet. A1

**(ii)** idea of $\Delta\phi = ½v^2$ C1
$(6.8 - 5.3) \times 10^7 = ½v^2$
$v = 5.5 \times 10^3$ ms$^{-1}$ A1

***5.***

**(a)** angle (subtended) <u>at centre</u> of circle — B1
(by) arc equal in length to radius — B1

**(b) (i)** point S shown below C — B1

**(ii)** (max) force / tension = weight + centripetal force — C1
centripetal force = $mr\omega^2$ — C1
$15 = 3.0/9.8 \times 0.85 \times \omega^2$ — C1
$\omega = 7.6 \text{ rad s}^{-1}$ — A1

***6.***

**(a)** gravitational force provides the centripetal force — B1
$GMm/r^2 = mr\omega^2$ *(must be in terms of $\omega$)* — B1
$r^3\omega^2 = GM$ <u>and</u> $GM$ is a constant — B1

**(b) (i) 1.** for Phobos, $\omega = 2\pi/(7.65 \times 3600)$ — C1
$= 2.28 \times 10^{-4} \text{ rad s}^{-1}$
$(9.39 \times 10^6)^3 \times (2.28 \times 10^{-4})^2 = 6.67 \times 10^{-11} \times M$ — C1
$M = 6.46 \times 10^{23} \text{ kg}$ — A1

**2.** $(9.39 \times 10^6)^3 \times (2.28 \times 10^{-4})^2 = (1.99 \times 10^7)^3 \times \omega^2$ — C1
$\omega = 7.30 \times 10^{-5} \text{ rad s}^{-1}$ — C1
$T = 2\pi/\omega = 2\pi/(7.30 \times 10^{-5})$
$= 8.6 \times 10^4 \text{ s}$
$= 23.6$ hours — A1

**(ii)** *either* almost 'geostationary'
*or* satellite would take a long time to cross the sky — B1

***7.***

**(a) (i)** $F = R\cos\theta$ — M1
$W = R\sin\theta$ — M1
dividing, $W = F\tan\theta$ — A0
*(max. 1 if derivation to final line not shown)*

**(ii)** <u>provides</u> the centripetal force — B1

**(b)** *either* $F = mv^2/r$ and $W = mg$
*or* $v^2 = rg/\tan\theta$ — C1
$v^2 = (14 \times 10^{-2} \times 9.8)/\tan 28°$ — C1
$= 2.58$
$v = 1.6 \text{ m s}^{-1}$ — A1

## 8.

| | | |
|---|---|---|
| 1(a) | constant speed **or** constant magnitude of velocity | **B1** |
| | acceleration (always) perpendicular to velocity | **B1** |
| 1(b)(i) | $F = mv^2 / r$<br>**or**<br>$v = r\omega$ **and** $F = mr\omega^2$ | **C1** |
| | $F = 790 \times 94^2 / 318$<br>$= 22000$ N | **A1** |
| 1(b)(ii) | centripetal acceleration: same | **B1** |
| | maximum speed: greater | **B1** |
| | time taken for one lap of the track: greater | **B1** |

## 9.

| | | |
|---|---|---|
| 1(a) | acceleration perpendicular to velocity | **B1** |
| 1(b)(i) | decreases | **B1** |
| 1(b)(ii) | (acceleration of) $9.8\,\text{m s}^{-2}$ is caused by weight of car<br>**or**<br>centripetal force must be greater than weight of car | **B1** |
| | (acceleration > $9.8\,\text{m s}^{-2}$) requires contact <u>force</u> from track<br>**or**<br>(centripetal force > weight) requires contact <u>force</u> from track | **B1** |
| 1(c) | $\frac{1}{2}mv_Y^2 = \frac{1}{2}mv_X^2 - mgh$ | **C1** |
| | $a = v^2 / r$ | **C1** |
| | $v_Y^2 = 3.8^2 - 2 \times 9.81 \times 0.62$ so $v_Y = 1.5\,\text{m s}^{-1}$<br>$a = 1.5^2 / 0.31 = 7.3\,\text{m s}^{-2}$ (which is less than $9.8\,\text{m s}^{-2}$) so no | **A1** |
| | **or** | |
| | $v_Y = \sqrt{(9.81 \times 0.31)} = 1.74\,\text{m s}^{-1}$ so $v_X^2 = 1.74^2 + 2 \times 9.81 \times 0.62$<br>$v_X = 3.9\,\text{m s}^{-1}$ (which is greater than $3.8\,\text{m s}^{-1}$) so no | **(A1)** |
| 1(d) | acceleration is independent of mass so makes no difference<br>**or**<br>mass cancels in the equation so makes no difference | **B1** |

## 10.

| | | |
|---|---|---|
| 1(a) | angle (subtended at centre of circle) when arc length = radius | B1 |
| 1(b) | $\omega = 2\pi / T$ | C1 |
| | $= 2\pi / (1.0 \times 60 \times 60)$<br>$= 1.7 \times 10^{-3}\ \text{rad s}^{-1}$ | A1 |
| 1(c)(i) | angle $= 1.7 \times 10^{-3} \times 1400$<br>$= 2.4$ rad | A1 |
| 1(c)(ii) | $L$ = arc length / angle<br>$= 0.44 / 2.4$<br>**or**<br>$L = 0.44 \times (3600 / 1400) / 2\pi$ | C1 |
| | $L = 0.18$ m | A1 |
| 1(c)(iii) | $a = r\omega^2$ | C1 |
| | $= 0.18 \times (1.745 \times 10^{-3})^2$<br>$= 5.5 \times 10^{-7}\ \text{m s}^{-2}$ | A1 |
| 1(d) | centripetal acceleration is negligible compared with acceleration of free fall<br>**or**<br>numerical comparison establishing answer to **(c)(iii)** $\ll 9.81$ | B1 |
| | resultant force is negligible compared with weight (of modelling clay) (so variation is negligible)<br>**or**<br>force exerted by minute hand (approximately) equal (and opposite) to weight of modelling clay | B1 |

## 11.

| | | |
|---|---|---|
| 2(a) | horizontal force on sphere causes centripetal acceleration | B1 |
| | weight of sphere is (now) equal to vertical component of tension<br>**or**<br>horizontal and vertical components (of force) (now) combine to give greater tension (in spring) | B1 |
| | greater tension in spring so greater extension of spring | B1 |
| 2(b)(i) | $r = 10.8 \times \sin 27° = 4.9$ cm | A1 |
| 2(b)(ii) | $T \cos \theta = mg$<br>**or**<br>$T \cos \theta = W$ **and** $W = mg$ | C1 |
| | $T \cos 27° = 0.29 \times 9.81$ leading to $T = 3.2$ N | A1 |
| 2(b)(iii) | $\Delta T = 3.2 - (0.29 \times 9.81)$ | C1 |
| | $k = \Delta T / \Delta x$<br>$= [3.2 - (0.29 \times 9.81)] / [10.8 - 8.5]$<br>$= 0.15\ \text{N cm}^{-1}$ | A1 |
| 2(c)(i) | centripetal acceleration $= (T \sin \theta) / m$<br>$= (3.2 \times \sin 27°) / 0.29$ | C1 |
| | $= 5.0\ \text{m s}^{-2}$ | A1 |